

Practicum No.2**Date: 30/01/2026****Summary of a Successful CSE/AIML/DSCS/DS Entrepreneur****I. Practical Significance**

This practicum introduces students to the systematic study and analysis of the profile of a successful entrepreneur in the domains of Computer Science, Artificial Intelligence & Machine Learning, Data Science & Cyber Security. By examining the journeys of leading technology innovators such as Sundar Pichai (Google), Jensen Huang (NVIDIA), Jeff Dean (Google DeepMind), N. Chandrasekaran (TCS), Elon Musk (Tesla, SpaceX, Neuralink), or Sridhar Vembu (Zoho), students gain insight into how technical expertise, innovation, risk-taking, and strategic decision-making drive the growth of software-driven and AI-powered enterprises. Through this case study, learners identify key entrepreneurial traits, technology adoption strategies, and personal characteristics that align with CSE/AIML/DSCS/DS-based start-up ventures. This exercise strengthens students' analytical, research, and presentation skills while deepening their understanding of entrepreneurship in the computing and AI sectors.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a Pitch deck, project proposals for small/medium-scale startups logically and scientifically.
- a) **Interpret** competitive dynamics shaping CSE-focused high-tech and software-driven markets. (CDO).
- b) **Synthesize** insights on incubation support systems for computing, software-product, and AI-based startups. (CDO).
- c) **Innovate** strategic direction for new ventures in advanced CSE domains including generative AI, distributed systems, and intelligent automation. (CDO).
- d) **Craft** compelling investment narratives for AI, cybersecurity, cloud, blockchain, and software innovation startups. (CDO).

III. Relevant CO(s)

- **CO1. Identify** entrepreneurial traits and match them with suitable business opportunities.

IV. Practical Outcome (PrO)

- **PrO2– Present** a detailed profile summary of a successful CSE/AIML/DSCS/DS entrepreneur (e.g., Sundar Pichai - Google, Jensen Huang- NVIDIA, Jeff Dean -Google DeepMind, N. Chandrasekaran – TCS, Elon Musk – AI ventures, or Sridhar Vembu – Zoho).

V. Related ADO(s)

- a) **Communicate** effectively through written and oral presentations. (PDO).
- b) **Demonstrate** curiosity and analytical thinking in understanding entrepreneurial journeys. (CDO).
- c) **Exhibit** ethical respect for individual achievements and data authenticity. (ADO).

VI. Minimum Underpinning Theory

An entrepreneur is an individual who identifies an opportunity, takes calculated risks, and develops innovative solutions to create value in the market. In the field of electronics, electrical, and biomedical engineering, entrepreneurship often involves technological innovation, problem-solving, and leadership to bridge the gap between research and commercialization. A case study method allows students to understand real-world examples of successful entrepreneurs — their background, education, challenges, strategies, and achievements.

Key theoretical concepts include:

- **Entrepreneurial Traits:** Innovation, risk-taking, perseverance, vision, leadership, adaptability, and resilience.
- **Opportunity Recognition:** The ability to identify market needs and match them with feasible technological solutions.
- **Entrepreneurial Process:** Idea generation → Feasibility analysis → Product development → Market launch → Growth and scaling.
- **Technology Entrepreneurship:** Application of engineering knowledge to develop commercially viable products or services.

Understanding entrepreneurial traits helps students reflect on their own capabilities and inspires them to pursue viable business opportunities in the engineering domain.

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and report preparation	1 per team
2	Internet access	For background study and references	As required
3	Reference materials	Articles, interviews, company profiles	1 set
4	Presentation slides / chart paper	For summary and display	1 per team

IX. Safety Precautions

- Ensure** all data used is from credible and verifiable sources.
- Avoid** plagiarism; cite all references properly.
- Present** information objectively and respectfully.

X. Procedure

- Select** a successful entrepreneur from CSE/AIML/DSCS/DS fields.
- Collect** data from reliable sources — articles, interviews, company websites.
- Analyze** background, motivation, and key ventures.
- Identify** entrepreneurial traits demonstrated.
- Match** traits with specific business opportunities.

XI. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	Asus	Used for data collection	1	-
2.	Internet Access	Jio	Used to access entrepreneur profiles, articles	As required	-
3.	Reference Material	Google website	Info about Jensen Huang and NVIDIA innovation	1	-
4.	Documentation	MS Word	Used for typing and formatting practicum record	1	-

XII. Actual Procedure Followed

1. In this study, I focused on Jensen Huang, a successful entrepreneur in the CSE and artificial intelligence fields.
2. I gathered information about his education, career path, and key venture, NVIDIA, from reliable sources like the official company website, articles, and interviews. I carefully analyzed the data to understand his technical skills, leadership qualities, and innovative methods.
3. I identified his entrepreneurial traits, including vision, risk-taking, adaptability, and problem-solving skills. I examined the opportunities he saw in GPU computing and AI acceleration in detail.

XIII. Observations and Data

S. No.	Entrepreneur	Major Venture	Key Traits	Opportunity Identified
1	Jensen Huang	NVIDIA – GPU Computing	Visionary leadership, innovation	Identified GPUs as powerful parallel processors beyond graphics
2	Jensen Huang	NVIDIA – Artificial Intelligence Platforms	Strategic thinking, adaptability	Recognized early growth of AI and machine learning applications
3	Jensen Huang	NVIDIA – Data Center & Cloud Computing	Risk-taking, technical expertise	Demand for high-performance computing in enterprise
4	Jensen Huang	NVIDIA – CUDA Software Ecosystem	Innovation, problem-solving	Created a developer-friendly platform for AI and scientific computing
5	Jensen Huang	NVIDIA – Robotics and Edge AI	Leadership, adaptability	Growth of intelligent automation and robotics

XIV. Results/Observations

It was observed that Jensen Huang successfully transformed NVIDIA from a graphics-focused company into a global leader in artificial intelligence, data center computing, and advanced technologies.

His early recognition of GPUs as parallel computing devices enabled their use in AI, machine learning, data science, and high-performance computing. The development of platforms such as CUDA, AI frameworks, and autonomous vehicle solutions contributed significantly to NVIDIA's growth.

Continuous investment in research and innovation allowed the company to remain competitive in rapidly evolving technology markets. Strong leadership and strategic decision-making helped in expanding NVIDIA into multiple domains such as healthcare, robotics, cloud computing, and the metaverse.

XV. Interpretation of Results

The results indicate that entrepreneurial success in the CSE and AI domain depends on the ability to identify future technology trends and apply engineering knowledge effectively.

Jensen Huang's journey shows that innovation combined with long-term vision can create sustainable business growth. His adaptability to changing market demands and focus on building a strong technology ecosystem helped NVIDIA achieve global leadership.

XVI. Conclusions

It can be concluded that aspiring entrepreneurs should develop strong technical foundations and stay updated with emerging technologies such as artificial intelligence, machine learning, and cloud computing. Taking calculated risks, investing in research and development, and maintaining ethical practices are essential for long-term success.

XVII. Suggested Practicum Related Questions

1. **Analyze** how the entrepreneur's technical expertise in areas such as AI/ML, software engineering, cybersecurity, or data science contributed to the creation and scaling of their venture.
2. **Identify** key entrepreneurial traits—such as perseverance, adaptability, and visionary thinking—demonstrated by the chosen entrepreneur and explain how these traits influenced major business milestones.
3. **Evaluate** the role of early career experiences, academic background, and domain exposure in shaping the entrepreneur's problem-solving approach and startup direction.
4. **Examine** the strategic decisions made by the entrepreneur in product development, market positioning, or technology adoption, and assess their impact on the venture's growth.
5. **Assess** how the entrepreneur leveraged emerging technologies (AI, cloud, blockchain, IoT, big data) to create competitive advantages and differentiate their startup in the tech ecosystem.
6. **Discuss** how the entrepreneur built and managed high-performance teams, highlighting leadership styles, decision-making patterns, and talent acquisition strategies.
7. **Trace** the evolution of the entrepreneur's business model and analyze how customer feedback, market challenges, or technological shifts influenced necessary pivots.
8. **Interpret** how the entrepreneur balanced innovation with ethical considerations, including data privacy, algorithmic fairness, cybersecurity, and responsible AI practices.
9. **Evaluate** the resilience and crisis-management abilities of the entrepreneur by examining their response to failures, funding challenges, market competition, or product setbacks.
10. **Explain** how the entrepreneur's journey provides inspiration and practical insights for aspiring CSE/AIML/DSCS/DS founders, particularly in identifying opportunities and navigating the early startup ecosystem.

XVIII References / Suggestions for further reading

Hisrich, R.D., Peters, M.P., & Shepherd, D.A. *Entrepreneurship*. McGraw-Hill Education.

XIX Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Selection of relevant entrepreneur and data collection	3	
2	Identification and analysis of entrepreneurial traits	4	
3	Logical correlation with business opportunities	4	

Performance Indicators		Maximum Marks	Marks Obtained
4	Team collaboration and communication	3	
5	Ethical use of data and originality of report	4	
Product-related Skills: 12 Marks - 40%			
1	Clarity and structure of report presentation	3	
2	Depth of analysis and interpretation	3	
3	Accuracy of business opportunity linkage	3	
4	Quality of conclusions and insights	3	
5	Timely submission of report and presentation	3	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236
2		
3		
4		

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Signature with Date
Process-Related skills (18)	Product-Related skills (12)	Total (30)		
